Dakini CRM - Technical Architecture Report

Generated on Live Supabase Database & Layer 3 Application Codebase | Role-Based Access Control & Relational Schema Mapping

Part 1: Role-Based Access Control (RBAC) Governance Matrix

Dakini CRM enforces a strict multi-tenant role hierarchy: **Owner** (full organization control) → **Manager** (scoped to assigned sales team members) → **Salesperson** (restricted to owned/assigned records only).

CRM Module	Action	Owner	Manager	Salesperson (Rep)
Leads	View Leads	**Full Organization**	Team Records + Assigned	Own Assigned Leads Only
	Unassigned Leads Pool	**Full Access**	Only if Authorized	**Strictly Forbidden**
	Create / Ingest	**Allowed**	Allowed	Allowed
	Edit / Status Change	**Allowed**	Team Leads Only	Own Assigned Leads
	Assign / Reassign	**To Anyone in Org**	Within Own Team Only	**Forbidden (403)**
	Delete (Soft Delete)	**Allowed**	Team Leads Only	**Forbidden (403)**
Deals / Opportunities	View Deals	**Full Organization**	Team Deals Only	Own Deals Only
	Stage Progression	**Any Deal in Org**	Team Deals Only	Own Deals Only
	Discount Limits	**Unlimited (100%)**	Up to 30%	Up to 10%
	Approve Discounts	**Allowed**	Allowed	**Forbidden (403)**
	Delete Deals	**Allowed**	Team Deals Only	**Forbidden (403)**
Accounts & Contacts	View / Create / Edit	**Full Organization**	Full Organization	Full Organization
	Delete Accounts	**Allowed (Exclusive)**	**Forbidden (403)**	**Forbidden (403)**
	Delete Contacts	**Allowed**	Team Contacts Only	**Forbidden (403)**
Tasks & Activities	View Tasks	**Full Organization**	Team Members' Tasks	Own Tasks Only
	Assign Tasks	**To Anyone in Org**	To Team Reps Only	To Self Only
	Delete Tasks	**Allowed**	Task Creator Only	Task Creator Only
Call Transcripts & AI	View & Analyze Calls	**Full Organization**	Team Calls Only	Own Calls Only
	Delete Call Analysis	**Allowed (Database Purge)**	Team Calls (DB Purge)	**Hidden / Forbidden (403)**
Campaigns	Create / Edit / Delete	**Allowed**	Allowed	**Forbidden (403)**
Sales Targets	Set Quotas / Targets	**To Managers & Reps**	To Team Reps Only	**Forbidden (403)**
Team Governance	Create Teams / Managers	**Allowed (Exclusive)**	**Forbidden (403)**	**Forbidden (403)**
	Manage Team Members	**Any Team**	Own Team Only	**Forbidden (403)**
Administration & Export	Create Users / Edit Roles	**Managers & Reps**	Reps in Team Only	**Forbidden (403)**
	Reset Passwords	**Any User in Org**	Team Members Only	**Forbidden (403)**
	Export Data (CSV/PDF)	**Full Organization**	Team Records Only	**Forbidden (403)**

Part 2: Relational Database Schema & Foreign Key Mapping

Complete schema mapping showing how tables connect across multi-tenancy, sales pipeline, activity tracking, AI analysis, invoicing, and auditing in Supabase PostgreSQL.

1. Multi-Tenancy, Identity & Governance Layer

Source Table	Target Table	Foreign Key Column	Type	Relationship / Behavior
users	**organizations**	organization_id	N:1	Tenant boundary isolation (CASCADE)
users	**roles**	role_id	N:1	RBAC role assignment (SET NULL)
users	**users**	manager_id	N:1	Self-referencing manager hierarchy (SET NULL)
roles	**organizations**	organization_id	N:1	Tenant scoped roles (CASCADE)
role_permissions	**roles**	role_id	N:1	Role to permission link (CASCADE)
role_permissions	**permissions**	permission_id	N:1	Permission definition link (CASCADE)
teams	**organizations**	organization_id	N:1	Sales team tenant scoping (CASCADE)
teams	**users**	manager_id	N:1	Team leader / manager assignment (SET NULL)
team_members	**teams**	team_id	N:1	Team membership link (CASCADE)
team_members	**users**	user_id	N:1	Salesperson member link (CASCADE)

2. Leads, Enrichment Signals & Marketing Campaigns

Source Table	Target Table	Foreign Key Column	Type	Relationship / Behavior
leads	**organizations**	organization_id	N:1	Tenant scoping boundary (CASCADE)
leads	**users**	assigned_to	N:1	Assigned salesperson / owner (SET NULL)
leads	**campaigns**	campaign_id	N:1	Attribution marketing campaign (SET NULL)
leads	**products**	product_id	N:1	Interested product catalog entry (SET NULL)
lead_signals	**leads**	lead_id	N:1	AI activity signal event tracking (CASCADE)
lead_conversions	**leads**	lead_id	1:1	Unique conversion record history (CASCADE)
lead_conversions	**accounts**	account_id	N:1	Account generated from lead conversion (CASCADE)
lead_conversions	**opportunities**	opportunity_id	N:1	Deal generated from lead conversion (SET NULL)
lead_conversions	**contacts**	contact_id	N:1	Contact generated from lead conversion (SET NULL)
campaign_members	**campaigns**	campaign_id	N:1	Campaign audience target list (CASCADE)
campaign_members	**leads**	lead_id	N:1	Lead associated with campaign (CASCADE)
campaign_members	**contacts**	contact_id	N:1	Contact associated with campaign (CASCADE)
campaign_responses	**campaign_members**	campaign_member_id	N:1	Member interaction / click response (CASCADE)

3. Accounts, Contacts & Sales Opportunities (Deals)

Source Table	Target Table	Foreign Key Column	Type	Relationship / Behavior
accounts	**organizations**	organization_id	N:1	Tenant boundary isolation (CASCADE)
accounts	**users**	owner_id	N:1	Account executive / owner (SET NULL)
contacts	**organizations**	organization_id	N:1	Tenant boundary isolation (CASCADE)
contacts	**accounts**	account_id	N:1	Company account that contact belongs to (CASCADE)
contacts	**users**	owner_id	N:1	Contact record owner (SET NULL)
deals	**organizations**	organization_id	N:1	Tenant boundary isolation (CASCADE)

Source Table	Target Table	Foreign Key Column	Type	Relationship / Behavior
deals	**accounts**	account_id	N:1	Primary client account for opportunity (CASCADE)
deals	**contacts**	contact_id	N:1	Primary decision-maker contact (SET NULL)
deals	**users**	owner_id	N:1	Deal owner / sales representative (SET NULL)
deals	**campaigns**	campaign_id	N:1	Source campaign for deal attribution (SET NULL)
deal_products	**deals**	deal_id	N:1	Deal items line specification (CASCADE)
deal_products	**products**	product_id	N:1	Catalog product line item (RESTRICT)

4. Activities, Call Transcripts & AI Intelligence Layer

Source Table	Target Table	Foreign Key Column	Type	Relationship / Behavior
activities	**organizations**	organization_id	N:1	Tenant boundary isolation (CASCADE)
activities	**leads**	lead_id	N:1	Activity related to unconverted lead (SET NULL)
activities	**accounts**	account_id	N:1	Activity related to client company (CASCADE)
activities	**contacts**	contact_id	N:1	Activity related to person contact (CASCADE)
activities	**deals**	opportunity_id	N:1	Activity related to sales opportunity (CASCADE)
activities	**users**	user_id	N:1	Salesperson creator / performer (SET NULL)
activity_participants	**activities**	activity_id	N:1	Meeting / call attendees (CASCADE)
activity_participants	**users**	user_id	N:1	Internal employee participant (CASCADE)
activity_participants	**contacts**	contact_id	N:1	External customer participant (CASCADE)
call_transcripts	**activities**	activity_id	1:1	Call transcript tied strictly 1:1 with Call Activity (CASCADE)
call_transcripts	**organizations**	organization_id	N:1	Tenant boundary isolation (CASCADE)
call_analysis	**call_transcripts**	transcript_id	N:1	AI analysis structured outputs (1:N for versioning, CASCADE)
call_analysis	**activities**	activity_id	N:1	Parent call activity link (SET NULL)
call_analysis	**leads**	lead_id	N:1	Direct lead association for sentiment analytics (SET NULL)
call_analysis	**contacts**	contact_id	N:1	Direct contact association (SET NULL)
call_analysis	**accounts**	account_id	N:1	Direct account association (CASCADE)
call_analysis	**users**	analyzed_by	N:1	Sales representative / analyst identity (SET NULL)
notes	**organizations**	organization_id	N:1	Tenant boundary isolation (CASCADE)
notes	**users**	user_id	N:1	Note author / salesperson (SET NULL)

5. Sales Targets, Invoices, Documents & Audit Trails

Source Table	Target Table	Foreign Key Column	Type	Relationship / Behavior
sales_targets	**organizations**	organization_id	N:1	Tenant boundary isolation (CASCADE)
sales_targets	**users**	assigned_to_user_id	N:1	Salesperson holding quota target (CASCADE)
sales_targets	**users**	assigned_by_user_id	N:1	Manager or Owner who set target (RESTRICT)
sales_targets	**teams**	team_id	N:1	Associated sales team quota (CASCADE)
invoices	**accounts**	account_id	N:1	Billed enterprise account (CASCADE)
invoices	**deals**	deal_id	N:1	Deal associated with invoice (SET NULL)
invoice_items	**invoices**	invoice_id	N:1	Line item within invoice (CASCADE)
invoice_items	**products**	product_id	N:1	Catalog product line item (SET NULL)
payments	**invoices**	invoice_id	N:1	Received payment transaction (CASCADE)
documents	**users**	uploaded_by	N:1	File uploader identity (SET NULL)

Source Table	Target Table	Foreign Key Column	Type	Relationship / Behavior
document_relations	**documents**	document_id	N:1	Document attachment to lead/deal/account (CASCADE)
document_versions	**documents**	document_id	N:1	File versioning history (CASCADE)
audit_logs	**organizations**	organization_id	N:1	Tenant boundary isolation (CASCADE)
audit_logs	**users**	user_id	N:1	Actor who performed the CRM operation (SET NULL)